Comparative Research on GPU, its Operating Systems and their Application in AWS Cloud Platforms

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Module Code: CN7029

Module Title: Computer Architecture and Networks

Table of Contents

1. Introduction 3

2. Task 1: GPU Operating System Implementation (Linux) 4

2.1 Show GPU and Graphics Information 5

2.2 Check Kernel Messages Related to GPU 5

2.3 Show GPU Memory and Information 6

2.4 Check GPU Devices in /dev/dri 6

2.5 Check GPU-Related Loaded Kernel Modules 7

2.6 Summary of Task 1 8

3. Task 2: AWS Cloud Platform Implementation 8

3.1 EC2 Launch Page 8

3.2 EC2 Instance Configuration 9

3.3 Instance Type Selection 9

3.4 Key Pair Creation 10

3.5 Network Configuration 10

3.6 Storage Configuration 11

3.7 EC2 Instance Launch Status 11

3.8 Monitoring Configuration 12

3.9 Stopping and Terminating Instances 13

3.10 Summary of Task 2 15

4. Task 3: Evaluation (Reflective Analysis) 15

4.1 Evaluation of GPU Command-Line Tools 16

4.2 Analysis of GPU Monitoring in Virtualized Environments 16

4.3 Evaluation of AWS Cloud Implementation 17

4.4 Comparison of Linux GPU Commands and Cloud Monitoring 17

4.5 Summary of Task 3 17
5. Task 4: Conclusion 18
5.1 Summary of Work 18
5.2 Key Findings 18
5.3 Contribution to Knowledge and Skills 19
5.4 Final Remarks 19
Reference 20

List of Figures

Figure 1 Show GPU and graphics info 5
Figure 2 Check kernel messages related to GPU 6
Figure 3 Show GPU memory and info 6
Figure 4 Check GPU devices 7
Figure 5 Check GPU-related loaded kernel modules 7
Figure 6 Ec2 launch page 8
Figure 7 Ec2 config 9
Figure 8 Instance type 10
Figure 9 Key pair creation 10
Figure 10 Network setting 11
Figure 11 Storage config 11
Figure 12 Ec2 instance launch status 12
Figure 13 Monitoring config 13
Figure 14 Stopping instance 14
Figure 15 Ec2 instance deletion 15

1. Introduction

Over the past few years, HPC has been an ever-growing field where GPUs have become popular in doing all kinds of workloads: scientific computing, machine learning, artificial intelligence, and cloud computing. GPUs are dedicated processors to execute parallel computations well with significant improvement over traditional Computer Processing Units (CPUs) with specific operations. The increase of the number of systems based on GPU has been accompanied by the emergence of cloud services, including Amazon Web Services (AWS) and Microsoft Azure, that can offer access to the scalable amounts of GNU without the necessity of users holding physical devices.

This coursework aims to do a comparative study to determine the use of GPU with benchmarking on Linux GPU operating systems command-line tools to monitor and diagnose. This research will also include a practical application, which will be based on the AWS cloud computing services, that is, the deployment and management of EC2 instances. This method incorporates both theoretical studies and practical experimentation to show how the use of GPU monitoring, managing resources, and deploying on a cloud can actually be a part of a contemporary computing environment.

The coursework will include four pieces, with the first task being a working session on the commands of the graphics processing unit operating system in Linux, the second task will include the practical work on AWS, and the third task will contain the reflection on the results presented in the preceding work. The concluding part will be included in the fourth task, and it will point out the main results and observations made. The report will supply an in-depth participant of the functionality of GPUs and their use under cloud computing conditions, as well as show the real skills in system arrangement, checking, and implementation.

2. Task 1: GPU Operating System Implementation (Linux)

[redacted]

```
┌──(kali㉿kali)-[~]
└─$ # U2835044 - Command 1: List GPU devices
lspci | grep -i "vga\|3d\|2d"

00:02.0 VGA compatible controller: VMware SVGA II Adapter
```

```
┌──(kali㉿kali)-[~]
└─$ # U2835044 - Command 2: GPU kernel logs
dmesg | grep -i drm

[    2.431457] ACPI: bus type drm_connector registered
[    2.848678] vmwgfx 0000:00:02.0: [drm] FIFO at 0x00000000f0000000 size is 2048 KiB
[    2.848693] vmwgfx 0000:00:02.0: [drm] VRAM at 0x00000000e0000000 size is 16384 KiB
[    2.848703] vmwgfx 0000:00:02.0: [drm] Running on SVGA version 2.
[    2.848710] vmwgfx 0000:00:02.0: [drm] Capabilities: rect copy, cursor, cursor bypass, cursor bypass 2, alpha cursor, extended fifo, pitchlock, irq mask
, gmr, traces, gmr2, screen object 2, command buffers, command buffers 2, gbobject,
[    2.848713] vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
[    2.848720] vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
[    2.848724] vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
[    2.848728] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[    2.848836] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[    2.848840] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 KiB, max mob pages = 4096
[    2.848842] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[    2.848844] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[    2.848845] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[    2.852076] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[    2.852379] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap 0x00000355
[    2.853055] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[    2.853075] vmwgfx 0000:00:02.0: [drm] Available shader model: Legacy.
[    2.854538] [drm] Initialized vmwgfx 2.21.0 for 0000:00:02.0 on minor 0
[    2.857699] fbcon: vmwgfxdrmfb (fb0) is primary device
[    2.862818] vmwgfx 0000:00:02.0: [drm] fb0: vmwgfxdrmfb frame buffer device
[    5.305706] systemd[1]: modprobe@drm.service - Load Kernel Module drm was skipped because of an unmet condition check (ConditionKernelModuleLoaded=!drm)
.
[    8.177453] 10:13:08.129838 main     Executable: /usr/bin/VBoxDRMClient
[    8.186089] 10:13:08.139141 main     VBoxDRMClient: found compatible device: /dev/dri/renderD128
[    8.191180] 10:13:08.144040 main     VBoxDRMClient: starting deferred resize events delivery
[    8.194203] 10:13:08.147035 main     VBoxDRMClient: IPC server socket access granted to all users
[    8.443425] 10:13:08.396174 Timer    VBoxDRMClient: push screen layout data of 1 display(s) to DRM stack, fPartialLayout=false, rc=VERR_INV
[    8.444726] 10:13:08.397460 Timer    VBoxDRMClient: Error: Failed to push display change as requested by host, rc=VERR_INVALID_PARAMETER
[  301.978838] 10:18:02.078579 IpcCLT-1012 VBoxDRMClient: IPC client connection started
[  304.497497] 10:18:04.598027 IpcCLT-1012 VBoxDRMClient: thread IpcCLT-1012 notified IPC Client that display 0 is now primary, rc=VINF_SUCCES

┌──(kali㉿kali)-[~]
└─$
```

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```
┌──(kali㉿kali)-[~]
└─$ # U2835044 - Command 3: Detailed GPU info
lspci -v -s 00:02.0

00:02.0 VGA compatible controller: VMware SVGA II Adapter (prog-if 00 [VGA controller])
        Subsystem: VMware SVGA II Adapter
        Flags: bus master, fast devsel, latency 64, IRQ 18
        I/O ports at d010 [size=16]
        Memory at e0000000 (32-bit, prefetchable) [size=16M]
        Memory at f0000000 (32-bit, non-prefetchable) [size=2M]
        Expansion ROM at 000c0000 [virtual] [disabled] [size=128K]
        Kernel driver in use: vmwgfx
        Kernel modules: vmwgfx


┌──(kali㉿kali)-[~]
└─$
```

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```
┌──(kali㉿kali)-[~]
└─$ # U2835044 - Command 4: GPU devices in /dev/dri
ls -l /dev/dri/

total 0
drwxr-xr-x  2 root root         80 Jan  3 15:43 by-path
crw-rw----+ 1 root video  226,   0 Jan  3 15:48 card0
crw-rw----+ 1 root render 226, 128 Jan  3 15:48 renderD128

┌──(kali㉿kali)-[~]
└─$
```

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```
┌──(kali㉿kali)-[~]
└─$ # U2835044 - Command 5: Loaded GPU kernel modules
lsmod | grep -i "drm\|vmwgfx"

vmwgfx                 491520  2
drm_ttm_helper          16384  2 vmwgfx
ttm                    135168  2 vmwgfx,drm_ttm_helper
drm_client_lib          12288  1 vmwgfx
drm_kms_helper         258048  3 vmwgfx,drm_ttm_helper,drm_client_lib
drm                    831488  8 vmwgfx,drm_kms_helper,drm_ttm_helper,drm_client_lib,ttm

┌──(kali㉿kali)-[~]
└─$
```

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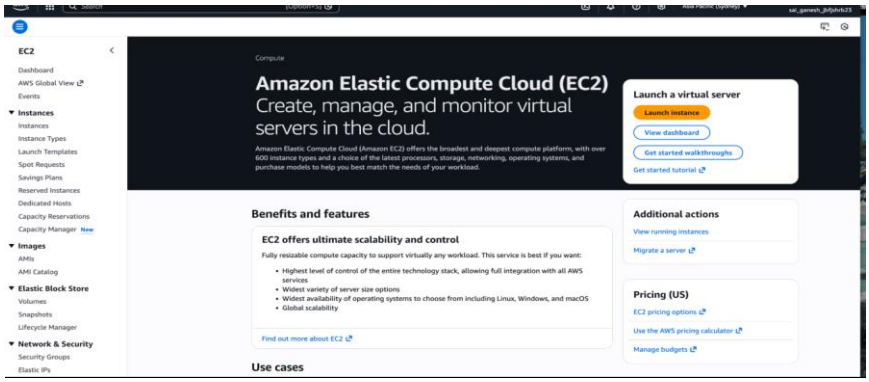

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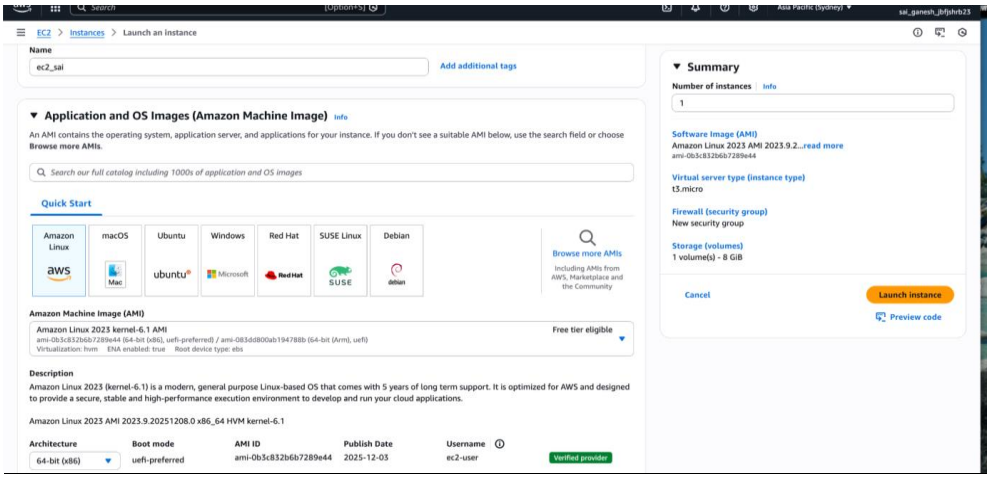

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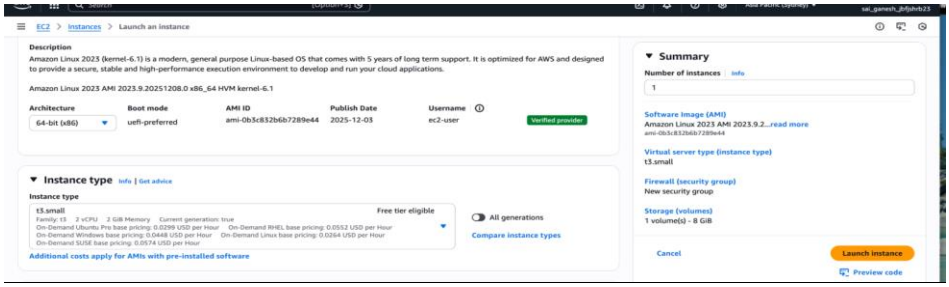

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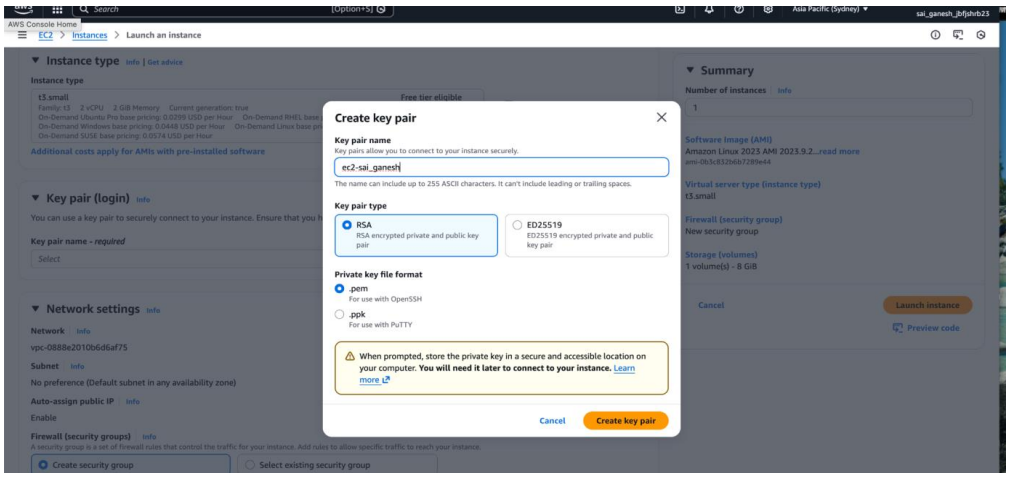

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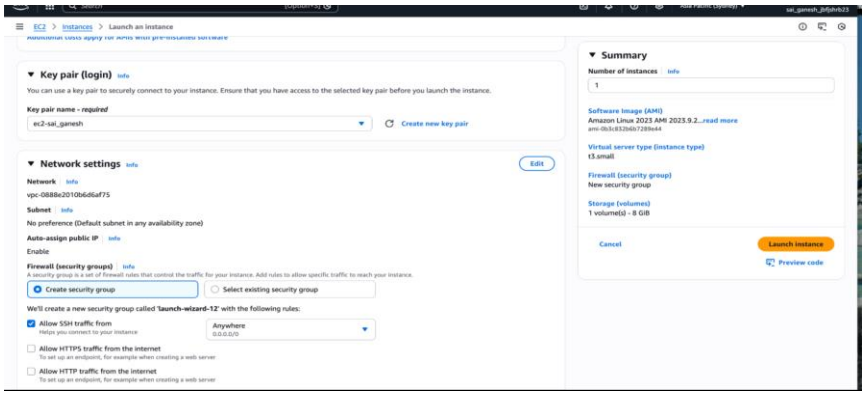

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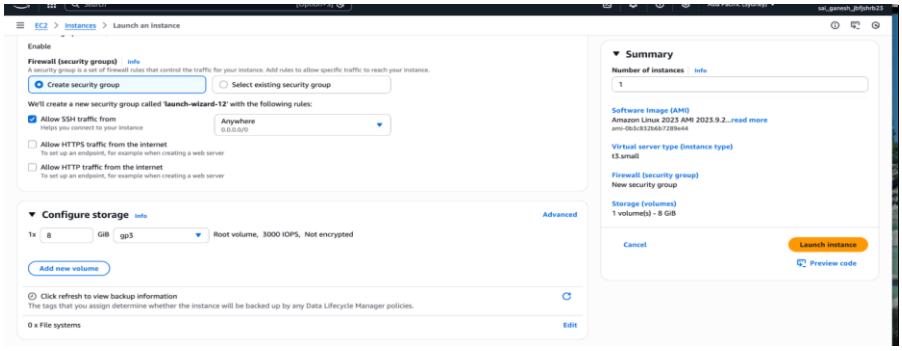

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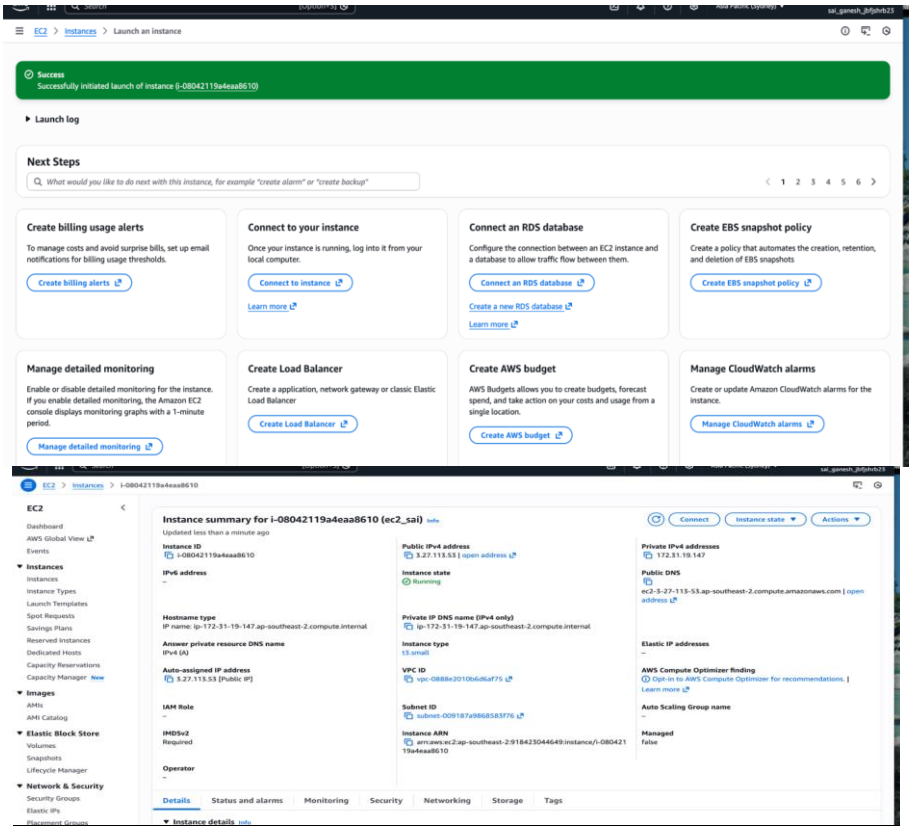

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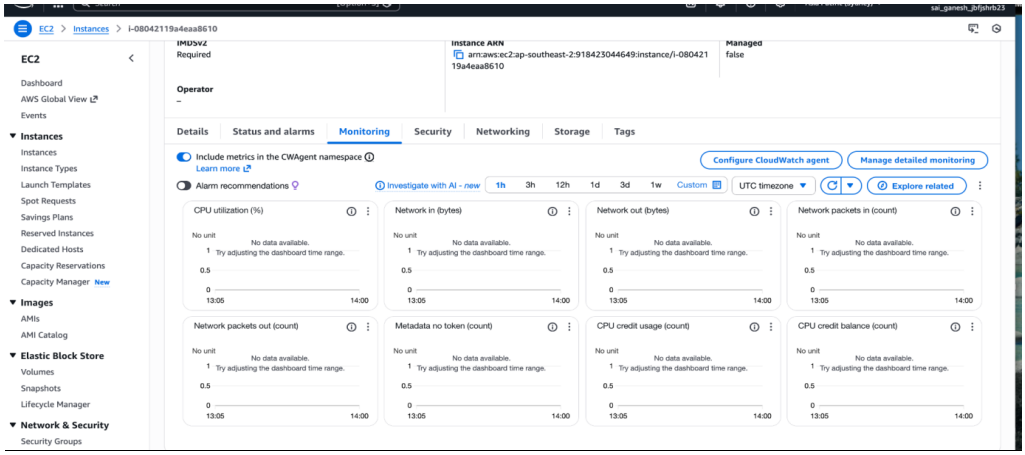

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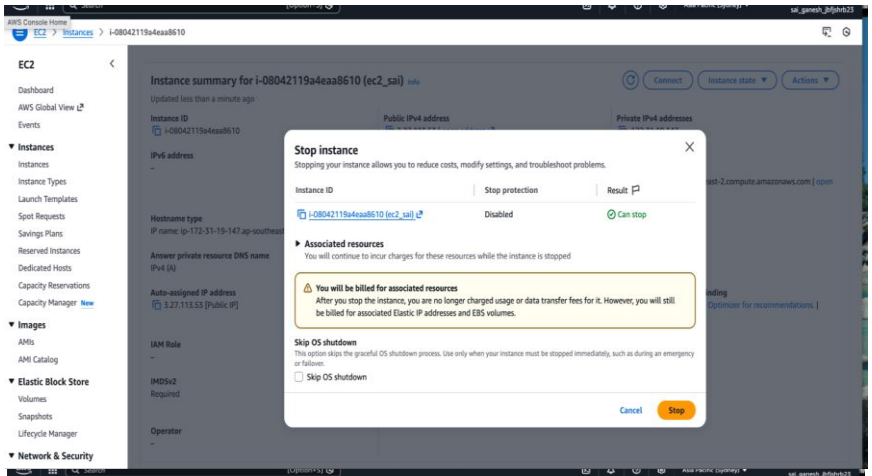
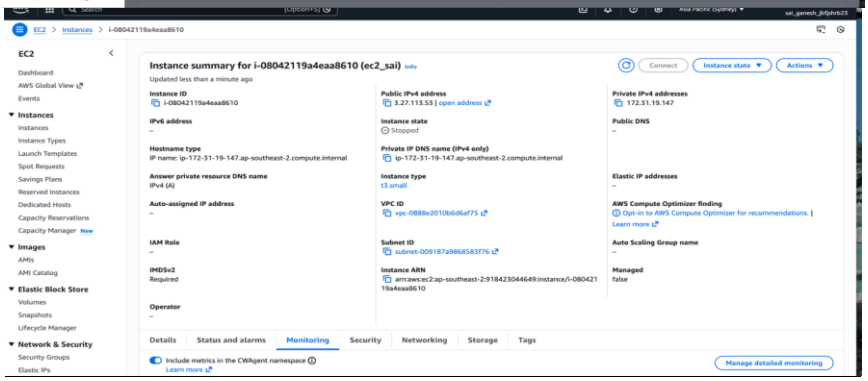

AWS Console Home
EC2 > Instances > i-08042119a4eaa8610
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Instance summary for i-08042119a4eaa8610 (ec2_sai) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-08042119a4eaa8610
IPv6 address
Hostname type
Answer private resource DNS name
IPv4 (A)
Auto-assigned IP address
3.27.113.53 [Public IP]
IAM Role
IMDSv2
Required
Operator
Public IPv4 addresses
Stop instance
Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.
Instance ID
Stop protection
Result
i-08042119a4eaa8610 (ec2_sai)
Disabled
Can stop
Associated resources
You will continue to incur charges for these resources while the instance is stopped
You will be billed for associated resources
After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.
Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.
Skip OS shutdown
Cancel
Stop


EC2 > Instances > i-08042119a4eaa8610
EC2
Instance summary for i-08042119a4eaa8610 (ec2_sai) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-08042119a4eaa8610
IPv6 address
Hostname type
IP name: ip-172-31-19-147.ap-southeast-2.compute.internal
Answer private resource DNS name
IPv4 (A)
Auto-assigned IP address
IAM Role
IMDSv2
Required
Operator
Public IPv4 address
3.27.113.53 | open address
Instance state
Stopped
Private IP DNS name (IPv4 only)
ip-172-31-19-147.ap-southeast-2.compute.internal
Instance type
t3.small
VPC ID
vpc-0888e2010b6d6af75
Subnet ID
subnet-009187a9868583f76
Instance ARN
arn:aws:ec2:ap-southeast-2:918423044649:instance/i-08042119a4eaa8610
Private IPv4 addresses
172.31.19.147
Public DNS
Elastic IP addresses
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
Auto Scaling Group name
Managed
false
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Include metrics in the CWAgent namespace
Learn more
Manage detailed monitoring

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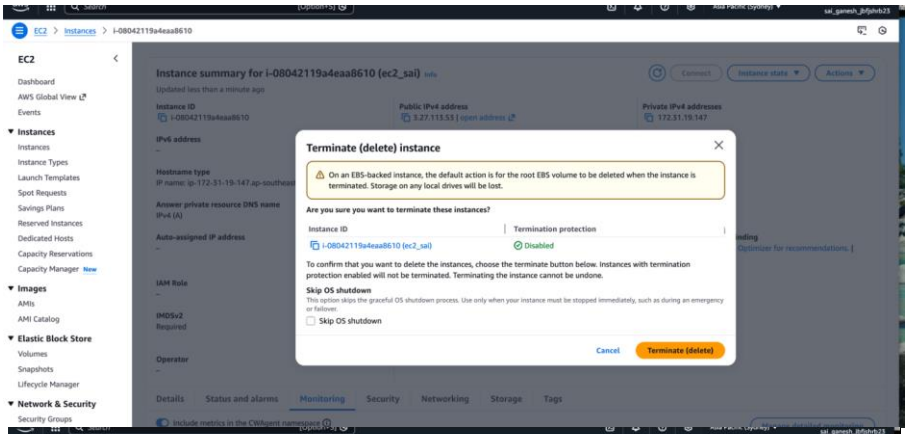

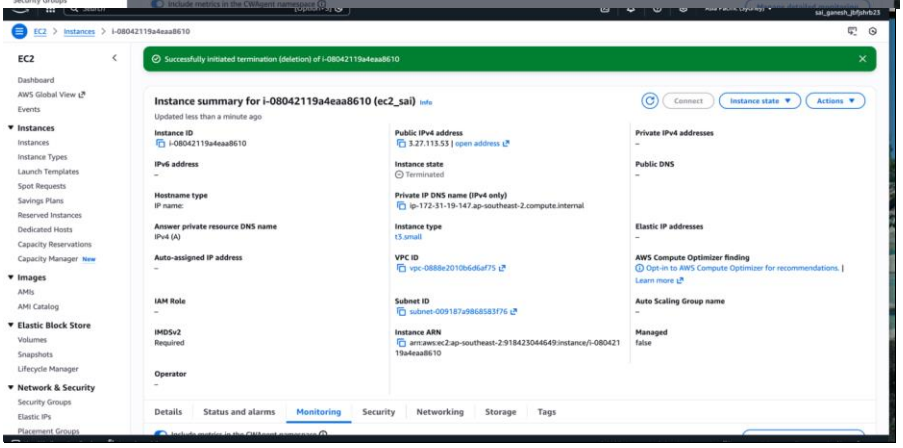

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